

Extremal rotating black holes in five-dimensional Einstein–Gauss–Bonnet gravity, constructed directly at zero temperature

BHs project, `work/kkr-extremal`

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Abstract

We construct the extremal (zero-temperature) black holes of five-dimensional Einstein–Gauss–Bonnet gravity that carry two equal angular momenta, directly at $T = 0$ and not as a limit of hot solutions. The construction follows the radial gauge of Kleihaus, Kunz and Radu [1], in which the degenerate horizon sits at the origin. We combine it with a spectral collocation method in which the mass is read as a *value* at infinity rather than as a derivative. Two hidden zero modes of that gauge had to be identified and removed before the problem became well posed. With these solutions we reproduce the extremal boundary of the domain of existence in figure 1 of [1], to the accuracy with which the published curves can be read. We then compute the extremal mass $M_{\text{ext}}(J, \alpha)$ and its slope $\partial M_{\text{ext}}/\partial \alpha|_J$ at fixed angular momentum, the slope directly at $T = 0$ by implicit differentiation of the discrete equations. These agree with the zero-temperature extrapolation of non-extremal solutions in this project’s manuscript to a few parts in 10^6 . That is far better than the uncertainty the manuscript attaches to its own slope. Along the way we find that the extremal horizon is not smooth at finite coupling: a perturbation of the near-horizon geometry grows with a non-integer power of the distance to the horizon. This rules out an exact power-series description of the solutions at the horizon.

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1 Introduction

In Einstein gravity a rotating black hole cannot carry an arbitrary angular momentum at a given mass. For the five-dimensional Myers–Perry solution with two equal angular momenta $J_1 = J_2 = J$ the bound is $M \geq \frac{3}{2}\pi^{1/3}J^{2/3}$ (in units $G = 1$). The black hole that saturates it is *extremal*: its Hawking temperature vanishes and its horizon becomes degenerate. Adding a Gauss–Bonnet term to the action,

$$I = \frac{1}{16\pi} \int d^5x \sqrt{-g} (R + \alpha L_{\text{GB}}), \quad L_{\text{GB}} = R^2 - 4R_{\mu\nu}R^{\mu\nu} + R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}, \quad (1)$$

moves that bound. The question this project's manuscript asks is how far, and in which direction, once α is no longer small. To first order the answer is known [3]:

$$M_{\text{ext}} = \frac{3}{2}\pi^{1/3}J^{2/3} + \pi\alpha + \mathcal{O}(\alpha^2). \quad (2)$$

There are two ways to reach the extremal solutions at finite coupling. The manuscript follows non-extremal black holes towards low temperature at fixed α and extrapolates to $T = 0$. It obtains the slope $\partial M_{\text{ext}}/\partial\alpha|_J$ as the zero-temperature limit of a thermodynamic potential Ψ conjugate to α . Kleihaus, Kunz and Radu (KKR below) [1] instead built the extremal black holes *directly*, in a radial gauge that puts the degenerate horizon at $r = 0$. They used them to map the domain of existence of the family. They published curves in figure 1 of their paper, but no tables, no stated accuracy and no mass curve $M_{\text{ext}}(J, \alpha)$.

The aim of the work reported here was therefore threefold:

1. rebuild KKR's direct construction and check it against their published curves;
2. use it to obtain $M_{\text{ext}}(J, \alpha)$ and its slope at $T = 0$ *without any extrapolation in temperature*, and so test the manuscript's extrapolated numbers by an independent route;
3. understand why the direct construction is numerically hard. It took several failed attempts, and the reasons turned out to be physical.

The main results are the following. The extremal solutions were constructed for $0 \leq \alpha \leq 3.118$ in units of the horizon size, which covers $0 \leq x \leq 0.863$ in KKR's variable $x = 3\pi\alpha/4M$. KKR's extremal curves are reproduced to about 10^{-3} , the precision to which their figure can be read. At the manuscript's twelve couplings the extremal mass agrees with the manuscript's extrapolation to 6.2×10^{-6} and the slope to 1.9×10^{-5} . The slope starts at the perturbative value π , falls to 1.0710 and rises again. And the extremal horizon is only C^1 in the natural radial coordinate, because of a near-horizon mode with a non-integer exponent.

The rest of the report is organised as follows. Section 2 sets up the metric, the conserved quantities and the reduced variables. Section 3 studies the near-horizon geometry and finds the non-smooth mode. Section 4 describes the numerical method, including the attempts that failed and why. Section 5 lists the checks. Section 6 presents the results, and section 7 states what was not established. A short guide to reproducing everything is in appendix A.

2 The model

2.1 Action, ansatz and conventions

We use the action (1) with $G = 1$ and $\alpha > 0$. This is also the normalisation of KKR. Brihaye, Kleihaus, Kunz and Radu [2] write $\frac{\alpha}{4}L_{\text{GB}}$ instead, so their coupling is four times ours. When the two angular momenta are equal the isometry group is enlarged to $\mathbb{R}_t \times U(2)$. All angular dependence then factors out of the field equations and the problem reduces to ordinary differential equations in a radial coordinate. We write the metric as in [2]:

$$ds^2 = -b(r) dt^2 + \frac{dr^2}{f(r)} + g(r)[d\theta^2 + \sin^2 \theta \cos^2 \theta (d\varphi_1 - d\varphi_2)^2] + h(r)[\sin^2 \theta d\varphi_1 + \cos^2 \theta d\varphi_2 - w(r) dt]^2. \quad (3)$$

Here g is the size of the round part of the S^3 , h the size of the fibre (so $h \neq g$ measures the squashing of the horizon sphere caused by rotation), and w the rotation of the frame. The functions (b, f, g, h, w) are what we solve for. One of them can be fixed by choosing the radial coordinate (a *gauge*), and much of what follows is about making that choice well.

Field equations. Rather than evaluating the Einstein–Gauss–Bonnet tensor for every ansatz, we used the reduced one-dimensional Lagrangian of [2] (their eq. 2.11). Its Euler–Lagrange equations with respect to b, f, g, h, w are the field equations. The project already had this Lagrangian, transcribed and validated in the gauge $g = r^2$. The extremal construction needs other gauges, so the equations were re-derived here with g left free. They were then checked against an independent evaluation of the full tensor equations $(G + \alpha H)^{\mu\nu}$ on the five-dimensional metric. The check uses *generic* trial functions, not solutions, so it tests every term of the equations and not only their combination on shell. The two routes agree to 3×10^{-14} . The w -equation has a first integral, the conserved momentum of w , which gives the angular momentum at any radius:

$$J = -\frac{\pi}{16} \frac{\partial L}{\partial w'}, \quad (4)$$

where L is the reduced Lagrangian. It is the analogue of eq. (2.12) of [2].

2.2 Physical quantities and reduced variables

From a solution we read the mass M and the angular momentum J in each plane, the horizon angular velocity $\Omega_H = w(r_H)$, the horizon area A_H , and the entropy. The entropy is not $A_H/4$ in this theory. It carries a Gauss–Bonnet correction from the intrinsic curvature of the horizon [6], which for the squashed sphere gives

$$S = \frac{\pi^2}{2} g_H \sqrt{h_H} \left[1 + 2\alpha \left(\frac{8}{g_H} - \frac{2h_H}{g_H^2} \right) \right], \quad A_H = 2\pi^2 g_H \sqrt{h_H}. \quad (5)$$

The theory has one scale, α (a length squared), so only scale-free combinations matter. We use two sets of them. To compare with KKR we use their mass-normalised variables

$$x = \frac{3\pi\alpha}{4M}, \quad j = \left(\frac{3}{2}\right)^{3/2} \frac{\sqrt{\pi} J}{M^{3/2}}, \quad a_H = \frac{3}{32} \sqrt{\frac{3}{2\pi}} \frac{A_H}{M^{3/2}}, \quad s = \frac{3}{8} \sqrt{\frac{3}{2\pi}} \frac{S}{M^{3/2}}, \quad (6)$$

normalised so that the Schwarzschild–Tangherlini black hole has $a_H = s = 1$ and extremal Myers–Perry has $j = 1$. For the extremal mass at fixed angular momentum we use the variables of the manuscript:

$$y = \frac{\alpha}{J^{2/3}}, \quad \mu = \frac{M_{\text{ext}}}{J^{2/3}}, \quad M_{\text{ext}}(J, \alpha) = J^{2/3} \mu(y). \quad (7)$$

Because of scale invariance the extremal branch is a single curve $\mu(y)$. Its derivative is exactly the quantity we want: $\partial M_{\text{ext}}/\partial\alpha|_J = \mu'(y)$. At $\alpha = 0$, $\mu(0) = \frac{3}{2}\pi^{1/3} = 2.19689$, and by (2) $\mu'(0) = \pi$.

Thermodynamic identities at $T = 0$. With α treated as a thermodynamic variable [7], the first law and the Smarr relation (which follows from scaling: $M \sim L^2$, $S, J \sim L^3$, $\alpha \sim L^2$) read

$$dM = T dS + 2\Omega_H dJ + \Psi d\alpha, \quad 2M = 3TS + 6\Omega_H J + 2\Psi\alpha. \quad (8)$$

At $T = 0$ the first gives $\partial M_{\text{ext}}/\partial\alpha|_J = \Psi$ and $\partial M_{\text{ext}}/\partial J|_\alpha = 2\Omega_H$. Combining the second with (7) gives two identities. We do not impose them, so they serve as checks:

$$\Psi = \frac{M - 3\Omega_H J}{\alpha} \quad (\text{Smarr at } T = 0), \quad 3\Omega_H J^{1/3} = \mu - y\mu'. \quad (9)$$

(The second follows from differentiating $M_{\text{ext}} = J^{2/3}\mu(y)$ with respect to J and using $\partial M_{\text{ext}}/\partial J = 2\Omega_H$. We derived it here; it is also an equation of the manuscript.)

3 The extremal throat and a horizon that is not smooth

3.1 The near-horizon geometry

Close to a degenerate horizon the geometry of an extremal black hole approaches a simpler, exact solution: AdS_2 fibred over the squashed three-sphere. In the ansatz (3) it reads

$$b = v_1 \rho^2, \quad f = \rho^2/v_1, \quad g = G_0, \quad h = H_0, \quad w = \Omega_H + w_1 \rho, \quad (10)$$

with ρ the distance from the horizon and constant (v_1, G_0, H_0, w_1) . Put into the field equations, this reduces them to three algebraic relations. With the overall scale fixed by $G_0 = 1$, the solutions form a one-parameter family labelled by α . This is the attractor solution of Brihaye et al. [2], obtained there with Sen's entropy function [5] and quoted in closed form by KKR (their eqs. at7–at9). We solved the algebraic system numerically and reproduced KKR's closed forms to round-off on 60 points. There is one convention difference: the “ J ” of KKR's eq. (at9) is $J_1 + J_2$, twice the per-plane J used elsewhere in their paper and here. At $\alpha = 0$ the throat is that of extremal Myers–Perry, with $S = 2\pi J$.

3.2 Perturbations: a non-integer power

A full extremal black hole is not exactly the throat. Near the horizon it is the throat plus small corrections that grow with ρ . To find out what powers of ρ those corrections can contain, we perturbed (10):

$$b \rightarrow b(1 + \epsilon\beta\rho^\gamma), \quad g \rightarrow G_0(1 + \epsilon\eta\rho^\gamma), \quad h \rightarrow H_0(1 + \epsilon\chi\rho^\gamma), \quad w' \rightarrow w_1(1 + \epsilon\omega\rho^\gamma), \quad (11)$$

keeping f fixed (this fixes the radial coordinate). At first order in ϵ every equation becomes a single power of ρ times a linear form in $(\beta, \eta, \chi, \omega)$. A perturbation exists only when that linear system loses rank, which determines the allowed exponents γ . They come in the pairs $(\gamma, -1 - \gamma)$ familiar from fields in AdS_2 . The member with $\gamma > 0$ is the one that can appear in a black hole that is regular at the horizon.

At $\alpha = 0$ all exponents are integers, $\{-3, -2, 0, 0, 1, 2\}$, and the extremal Myers–Perry black hole is analytic in ρ . At finite coupling the growing exponent 2 moves off the integers (figure 1). It first decreases to a minimum $\gamma = 1.773$ at $y = 0.029$, then returns through 2 at $y = 0.71$ and through 3 at $y = 4.51$. We scanned up to $y = 36.3$, where it has grown beyond 5.

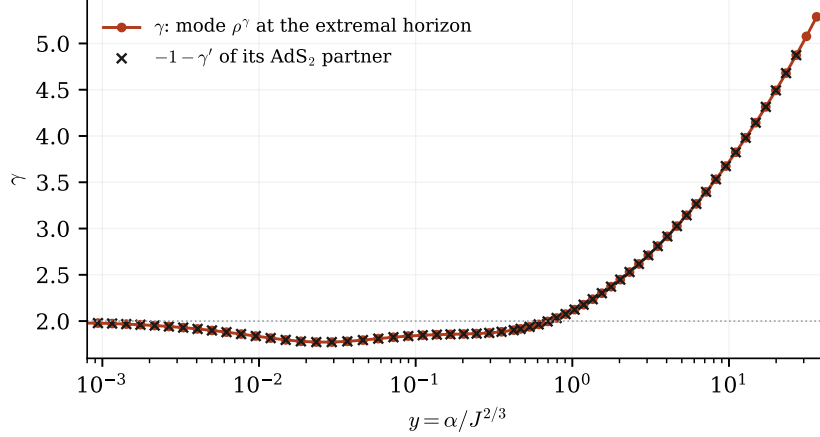


Figure 1: Exponent γ of the growing perturbation $\delta \propto \rho^\gamma$ of the extremal throat, as a function of $y = \alpha/J^{2/3}$ (dots and line). The crosses show $-1 - \gamma'$ for the AdS_2 partner exponent γ' . They fall on the same curve, which checks the pairing. Extremal Myers–Perry ($y = 0$) has $\gamma = 2$ exactly.

Why it matters. A generic extremal solution contains this mode with some non-zero amplitude, so near the horizon its metric functions behave like $c_0 + c_1\rho + c_\gamma\rho^\gamma + \dots$. For $\gamma < 2$, which is the case for $0 < y < 0.71$, they are once but not twice differentiable in ρ at the horizon. Two consequences follow:

- In KKR’s coordinate r one has $\rho \propto r^2$, so the solution contains $r^{2\gamma}$. Their remark that the solutions take “an approximate form as a power series in r ” near the horizon is therefore an approximation, never an exact statement at finite coupling.
- Spectral methods converge exponentially for analytic functions but only algebraically when such a term is present. Any quantity read from a high derivative is affected most. Section 4 shows how this decided the choice of unknowns.

The phenomenon is the five-dimensional, Gauss–Bonnet counterpart of the result of Horowitz, Kolanowski, Remmen and Santos [4]: higher-curvature corrections make the horizons of asymptotically flat extremal rotating black holes non-smooth. We have *not* computed curvature components in a parallelly propagated frame, so we make no claim about whether the horizon is singular in their sense. Our statement is about the differentiability of the metric functions.

4 The numerical method

4.1 What makes the extremal problem hard

A non-extremal horizon is a simple zero of b and f , and the standard approach of [2] (gauge $g = r^2$, horizon at $r = r_H$) handles it well. At an extremal horizon b and f have a double zero. The obvious extension is to keep $g = r^2$, factor $(r - r_H)^2$ out of b and f , and solve for the smooth remainders. *This does not work.* We tried it with Chebyshev collocation, and the linear systems became nearly singular as the resolution grew: condition numbers reached 10^{12} – 10^{13} and the smallest singular value tended to zero. The reason is that in this gauge the nearby *non-extremal* solutions are also almost representable. Moving slightly off extremality multiplies both b and f by the same factor $\sim 1/(x + \delta)$, which polynomials of high degree can approximate ever better. The discrete problem cannot decide whether it is extremal, and it becomes ill-conditioned. We abandoned this route.

4.2 KKR's gauge and its two hidden zero modes

KKR use a different gauge. With their parameter $a = 1$ and $P_i = e^{2F_i}$,

$$\begin{aligned} b &= \frac{P_1 r^4}{u^2 + 1}, & f &= \frac{r^2}{P_1 u}, & g &= P_2 u, & h &= P_2 P_3 u \left(1 + \frac{1}{u^2}\right), \\ w &= \frac{\sqrt{2}}{u^2 + 1} + W, & u &= r^2 + 1, \end{aligned} \quad (12)$$

with four unknown functions P_1, P_2, P_3, W . At $P_i = 1$, $W = 0$ this is exactly extremal Myers–Perry, with the degenerate horizon at $r = 0$. The essential feature is that the product

$$b f = \frac{r^6}{u(u^2 + 1)} \quad (13)$$

does not depend on the unknowns. A non-extremal solution would need b and f to have simple zeros at some $r_H > 0$, which this product forbids. So the representation excludes the neighbours that spoiled the previous attempt.

KKR's gauge alone did not give a well-posed problem either. We first solved it at fixed a and α and found that Newton's method converged to solutions that differed from run to run, and that the Jacobian had null vectors. These null vectors were smooth, the same at every resolution, and did not come from the discretisation. We identified them as two *exact* zero modes:

1. **Size.** The parameter a is not the size of the black hole. At fixed a and α the horizon can still be rescaled, so there is a one-parameter family of solutions. We remove it by fixing the size of the horizon sphere, $g_H = P_2(0) = 1$, which is the same unit $r_H = 1$ the manuscript uses.
2. **A leftover radial gauge.** The condition $b f = \text{fixed}$ does not fix r completely. The shift $r \rightarrow r + c\sqrt{b f}$ preserves it. This shift grows like r^3 near the horizon and tends to a constant shift at infinity. We remove it by demanding that P_2 have no $1/r$ term at infinity.

We also learned something from a failed fix. Replacing one collocation equation by the Hamiltonian constraint (the f -equation, which the gauge-fixed system does not impose) made the conditioning *worse*. The constraint is kept only as a diagnostic.

4.3 Coordinate, unknowns, and how the mass is read

Radial coordinate. We compactify KKR's r as $s = r/(1+r) \in [0, 1]$, with the horizon at $s = 0$ and infinity at $s = 1$. We first used $\xi = r^2/(1+r^2)$, which looks natural because the equations contain r^2 . But the leftover gauge mode r^3 becomes $\xi^{3/2}$, a non-smooth function, and convergence was slow. In s , both r^3 and the $1/r^n$ tails at infinity are smooth. The only non-smooth term left is the physical $r^{2\gamma}$ of section 3.

Unknowns. Instead of P_i themselves we solve for Q_i defined by

$$P_i = 1 + (1-s)^2 Q_i \quad (i = 1, 2, 3), \quad W = (1-s)^2 Q_4. \quad (14)$$

Because $(1-s)^2 \simeq 1/r^2$ at large r , this builds three conditions into the representation: asymptotic flatness, the absence of $1/r$ terms (which removes the leftover gauge mode), and, most importantly, that the mass is a *value*:

$$P_1 = 1 + \frac{2c_t}{r^2} + \dots, \quad c_t = \frac{1}{2} Q_1(s=1), \quad M = \frac{3\pi}{4} (1 - c_t). \quad (15)$$

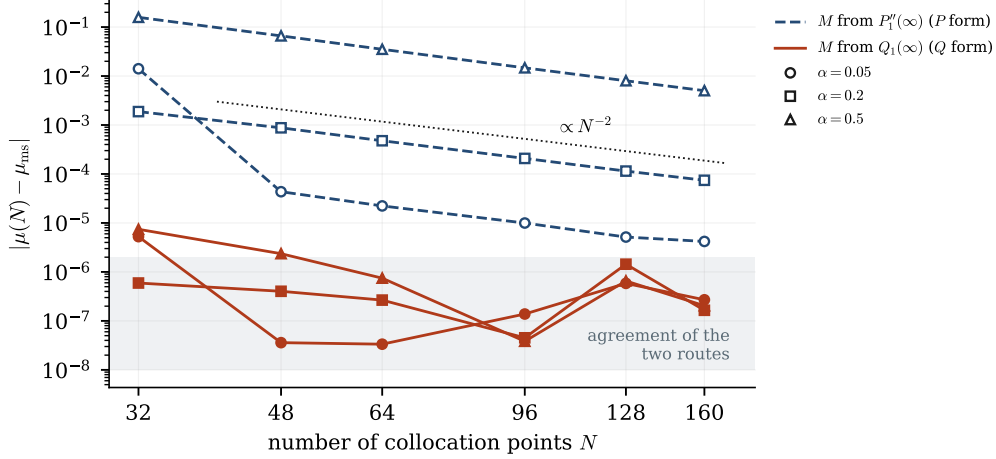


Figure 2: Convergence of the extremal mass with the number of collocation points N , at three couplings. The difference is taken from the manuscript’s independent $T \rightarrow 0$ value μ_{ms} . Dashed: mass from the second derivative of P_1 at infinity, which converges algebraically ($\propto N^{-2}$, fitted). Solid: mass as the value $Q_1(\infty)$, eq. (15). The shaded band is the level at which the two independent routes agree. Below it the comparison cannot discriminate.

In a first version we solved for P_i directly. The mass then came from the *second derivative* P_1'' at $s = 1$, which is sensitive to the slow, algebraic decay of the Chebyshev coefficients caused by $r^{2\gamma}$ at the other end of the domain. Figure 2 shows the effect. With P as unknowns the mass converges like N^{-2} and is wrong by 10^{-5} – 10^{-2} even at $N = 160$. With Q it reaches the 10^{-7} level already at $N \approx 48$. The angular momentum is read from the conserved momentum (4) in the interior of the domain, which avoids the fourth derivative its asymptotic tail would require. The horizon velocity is $\Omega_H = \frac{1}{\sqrt{2}} + W(0)$.

4.4 Discretisation and solution

The four equations for (P_1, P_2, P_3, W) are the variations of the gauge-fixed action. They are imposed at the N Chebyshev–Gauss–Lobatto points of $[0, 1]$, with the following changes at the boundaries:

- At the horizon ($s = 0$) the three P -equations are satisfied identically. We replace them by $dP_i/ds = 0$, which is the regularity condition (no odd power r^1). The W -equation is kept.
- At the first interior point the W -equation is replaced by the size condition $P_2(0) = 1$.
- At infinity ($s = 1$) all four equations are satisfied identically in the Q representation. They are imposed instead at a point midway between the last two nodes.

The residuals are long rational expressions. They were generated symbolically with SymPy and exported as NumPy code. To avoid catastrophic cancellation each is written in three algebraically equivalent forms: expanded about the horizon for $s \leq 0.12$, about infinity for $s \geq 0.88$, and unexpanded in between. The Jacobian is assembled analytically by the chain rule, with complex-step derivatives of the residuals. It agrees with a brute-force complex-step Jacobian of the whole discrete system to round-off. The nonlinear system is solved by Newton’s method with a backtracking line search.

4.5 Following the branch

At $\alpha = 0$ the exact solution is $Q_i = 0$ (extremal Myers–Perry). From there we increase α step by step. Each solution is computed at two resolutions, and a step is accepted only if all of the following hold:

- the residual is below 5×10^{-8} ;
- the Hamiltonian constraint, which is not imposed, is small;
- the last Chebyshev coefficients of every function are below 10^{-5} of the largest one, so the solution is resolved.

Each condition was added for a reason. The discrete system has *spurious roots*: solutions of the collocation equations whose Chebyshev coefficients do not decay and that violate the constraint at the level of 10^{-3} . Without the constraint and tail tests Newton sometimes jumps onto one of them. Near the round-off floor, Newton can also stop with grid-scale noise of order 10^{-6} in the highest modes. When the tails show this we drop the top quarter of the coefficients and solve again (a filtered restart). For $\alpha \gtrsim 1$ the step in α must be predicted along the tangent of the branch, $dQ/d\alpha$ (see below), or Newton loses the physical root.

The branch was computed in four phases. The constraint threshold was relaxed and the resolution raised as the solutions developed steeper structure near the horizon:

- constraint $< 10^{-6}$ and $N = 48/72$ up to $\alpha \approx 0.76$;
- constraint $< 10^{-5}$ and $N \geq 96/144$ up to $\alpha \approx 2.29$;
- constraint $< 10^{-4}$ and $N = 128/192$ up to $\alpha \approx 3.09$;
- finally N up to 312, ending at $\alpha = 3.118$.

In total 268 solutions were accepted. Going further was not possible in double precision. The constraint violation, concentrated next to the horizon, exceeded 10^{-4} even at $N = 312$. Raising N does not help, because the round-off floor of the residual grows with N . The best accuracy is reached at $N \approx 50$ –100. One earlier attempt at $N = 64/96$ was discarded because round-off noise made the step size collapse.

4.6 The slope without finite differences

The slope $\partial M_{\text{ext}}/\partial\alpha|_J$ could be obtained by differencing masses at neighbouring couplings, but that divides the error by the step. Instead we differentiate the discrete equations. If $R(Q; \alpha) = 0$ is the collocation system and $A = \partial R/\partial Q$ its Jacobian, then along the branch

$$A \frac{dQ}{d\alpha} = -\frac{\partial R}{\partial \alpha}, \quad (16)$$

which costs one linear solve with the matrix Newton already has. From $dQ/d\alpha$ we get $M_\alpha = dM/d\alpha = -\frac{3\pi}{8} dQ_1(1)/d\alpha$, and J_α by a complex-step derivative of (4). Along our branch the horizon size is fixed while α changes, so both M and J vary. Differentiating $M = J^{2/3}\mu(y)$ with $y = \alpha J^{-2/3}$ gives

$$\mu'(y) = \frac{M_\alpha - \frac{2}{3} (M/J) J_\alpha}{1 - \frac{2}{3} (\alpha/J) J_\alpha} = \left. \frac{\partial M_{\text{ext}}}{\partial \alpha} \right|_J. \quad (17)$$

The same tangent $dQ/d\alpha$ is the predictor used to follow the branch.

5 Checks

Every check listed here is evaluated in the code, and most are asserted (the script stops if they fail).

1. **Myers–Perry limit.** At $\alpha = 0$ the solver returns $M = 3\pi/4$, $J = \pi\sqrt{2}/4$, $\Omega_H = 1/\sqrt{2}$, $S = 2\pi J$, $j = 1$ and $a_H = s = \frac{1}{2}$, the exact extremal Myers–Perry values.
2. **Constraint.** The Hamiltonian constraint is never imposed. Over the whole branch it stays below 7.3×10^{-5} .
3. **Near-horizon data.** The horizon values of the full solutions must coincide with the throat of section 3 at the same α/g_H . With $\rho \propto r^2$ the identification is $v_1 = P_1(0)/4$ and $h_H = H_0$. They agree to 3.3×10^{-6} and 2.4×10^{-7} respectively.
4. **First law at $T = 0$.** $\Psi = M_\alpha - 2\Omega_H J_\alpha$ along the branch agrees with the slope (17) to 3.7×10^{-6} (median 7.2×10^{-8}). This is over the 183 branch points where the two resolutions agree to 10^{-5} .
5. **Smarr at $T = 0$.** $\Psi = (M - 3\Omega_H J)/\alpha$ agrees with the slope to 2.8×10^{-5} (median 5.4×10^{-7}) for $\alpha \geq 0.05$. At smaller α the division by α amplifies errors.
6. **Scaling identity.** $3\Omega_H J^{1/3} = \mu - y\mu'$ holds to 5.2×10^{-5} .
7. **Resolution.** Every quantity is computed at two (for the table, three) resolutions, and the differences are reported. These are differences, not error bounds.
8. **Independent re-run.** For this write-up the solver was run again from scratch: from extremal Myers–Perry, in steps of 0.01 up to $\alpha = 0.2$ at $N = 64$. It reproduced the extremal Myers–Perry values to 12 digits. At the six manuscript couplings in that range it matched the manuscript’s μ to 1×10^{-7} – 8×10^{-7} and its Ψ_{ext} to better than 2.2×10^{-5} (receipt in appendix A).

6 Results

6.1 Profiles of the extremal solutions

Figure 3 shows the four functions of KKR’s gauge, $F_i = \frac{1}{2} \ln P_i$ and W , for four couplings along the branch. All four vanish identically for extremal Myers–Perry, so the curves show the deviation caused by the Gauss–Bonnet term. The deviations are smooth and concentrated within a few horizon radii ($s \lesssim 0.7$, i.e. $r \lesssim 2.3$). They grow steadily with α , and so do their gradients near the horizon, which is why the resolution had to be raised along the branch. $F_2(0) = 0$ is the size condition $g_H = 1$. The value $F_3(0)$ measures the squashing of the horizon sphere, $h_H/g_H = 2e^{2F_3(0)}$. It becomes more negative with coupling: the fibre of the horizon, which for extremal Myers–Perry is elongated ($h_H/g_H = 2$), shrinks below the size of the round part ($h_H/g_H = 1.21, 0.74, 0.56$ at $\alpha = 0.05, 0.2, 0.5$).

6.2 Reproduction of KKR’s domain of existence

KKR plot the domain of existence of equal-spin black holes in the planes (x, a_H) , (x, s) and (x, t_H) , and against j in insets. Its lower boundary is the extremal set. To compare with our solutions we did not read the curves by eye. The figures in the arXiv source are gnuplot EPS files, in which every curve is a list of vertices in device coordinates. We extracted those vertices and mapped them to data coordinates using each plot’s own tick marks and labels. The main

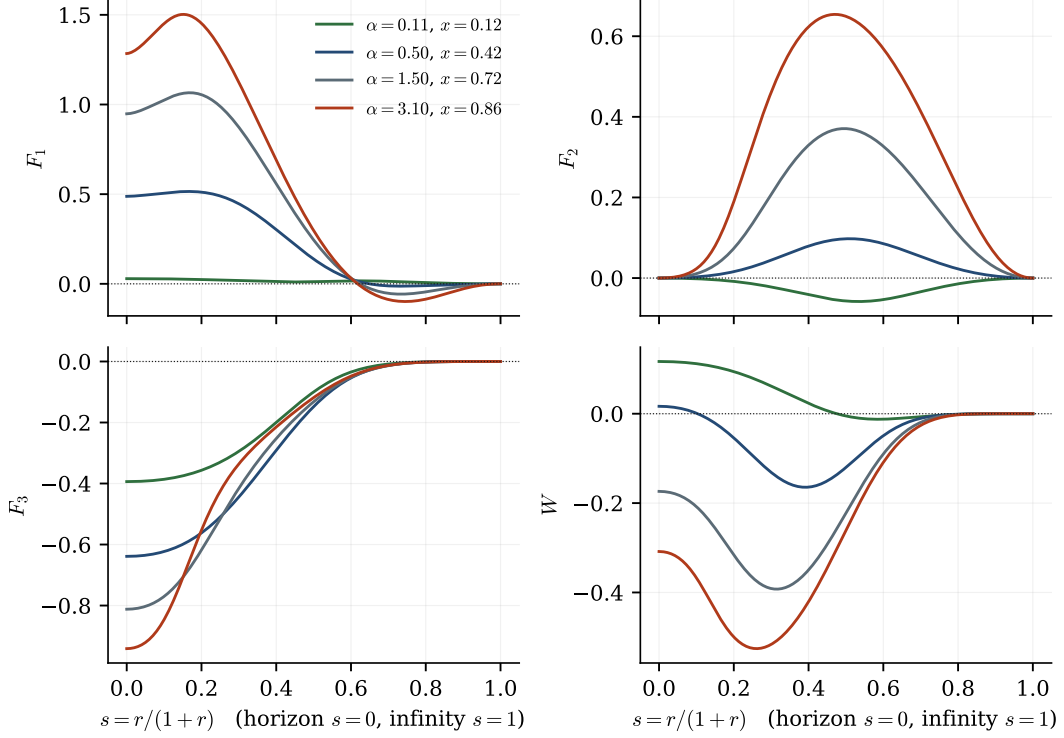


Figure 3: The functions $F_i = \frac{1}{2} \ln P_i$ and W of eq. (12) for extremal solutions at four couplings (units $g_H = 1$), against the compactified coordinate $s = r/(1+r)$. All vanish for extremal Myers–Perry.

plot and the inset were calibrated separately. Against the closed-form static curves, which are also drawn in the figure, this calibration is accurate to 3.7×10^{-4} , about one device unit.

Figure 4 shows the result. Our extremal branch (red) lies on KKR’s extremal curve (dashed blue) to 6.5×10^{-4} in $a_H(x)$ and 1.4×10^{-3} in $s(x)$. Against j the agreement is 1.2×10^{-3} and 2.6×10^{-3} . These numbers are at the level of the digitisation itself, so within what can be read from the figure the curves coincide. KKR’s bounds $j \leq 1$ and $x < 1$ hold on the whole branch, and j falls to 0.189 at the end.

KKR continue to $x \approx 0.92$. They draw a hand extrapolation from there to the singular static configuration at $(x, j) = (1, 0)$ and conjecture that the extremal set ends there. Our branch ends at $x = 0.863$, for the numerical reasons given in section 4, so we do not test that conjecture.

6.3 The extremal mass and its slope

Figure 5 is the main physics result. The top panel shows the extremal mass at fixed angular momentum, $\mu(y) = M_{\text{ext}}/J^{2/3}$, together with the first-order line $\mu(0) + \pi y$ of (2). The middle panel shows the slope $\mu'(y) = \partial M_{\text{ext}}/\partial \alpha|_J$ from (17). The slope starts at the perturbative value π and drops quickly, to a minimum of 1.0710 at $y = 0.195$. It then rises slowly again and reaches 2.1775 at the end of the branch ($y = 2.443$). It never changes sign. The extremal mass at fixed J therefore increases with the coupling throughout: the Gauss–Bonnet term tightens the extremality bound. Over the branch μ grows by 203%, well below the first-order line.

Comparison with the manuscript. The manuscript obtains μ and the slope Ψ_{ext} by fitting non-extremal solutions at low temperature and extrapolating to $T = 0$. It attaches two uncertainty estimates: the spread of three fits of different degree for μ , and a twenty-fit envelope for

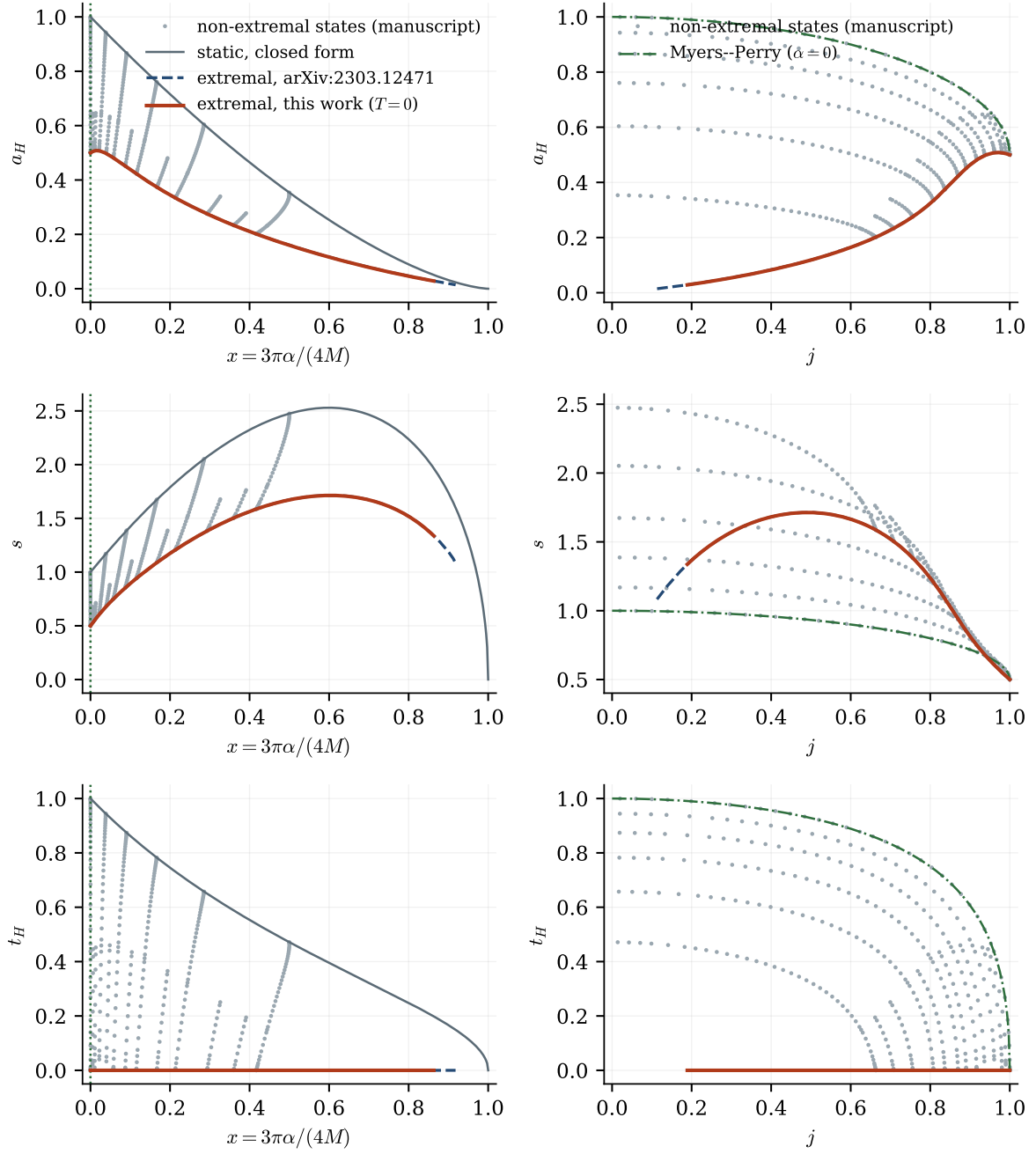


Figure 4: The equal-spin panels of figure 1 of KKR [1]. Left: against $x = 3\pi\alpha/4M$. Right: against j (their insets). Grey dots: non-extremal black holes from the manuscript. Solid grey: the static family (closed form). Dash-dotted green: Myers–Perry. Dashed blue: KKR’s extremal curves, extracted from their EPS files. Solid red: the extremal solutions constructed here at $T = 0$.

α	y	μ (here)	res.	μ (ms)	spread	μ' (here)	res.	Ψ_{ext} (ms)	envelope
0.005	0.004790	2.2115781	9.9×10^{-7}	2.2115792	1.3×10^{-3}	2.990940	5.4×10^{-6}	2.990943	1.1×10^{-1}
0.01	0.009845	2.2262777	7.1×10^{-7}	2.2262788	7.7×10^{-4}	2.824561	3.5×10^{-6}	2.824558	8.8×10^{-2}
0.02	0.020720	2.2550577	1.2×10^{-6}	2.2550579	1.6×10^{-3}	2.472846	4.4×10^{-6}	2.472827	1.6×10^{-1}
0.035	0.038427	2.2944442	9.4×10^{-7}	2.2944380	8.4×10^{-4}	2.000655	2.0×10^{-6}	2.000660	9.0×10^{-2}
0.05	0.056889	2.3280166	8.0×10^{-7}	2.3280172	5.7×10^{-6}	1.659079	9.0×10^{-7}	1.659080	8.6×10^{-3}
0.075	0.087750	2.3734822	2.2×10^{-7}	2.3734823	9.1×10^{-6}	1.325970	9.2×10^{-8}	1.325970	2.4×10^{-3}
0.1	0.117873	2.4107272	1.5×10^{-6}	2.4107287	3.7×10^{-5}	1.166215	1.9×10^{-8}	1.166214	8.3×10^{-4}
0.15	0.175201	2.4739927	4.7×10^{-7}	2.4739932	1.1×10^{-5}	1.071493	1.3×10^{-7}	1.071492	2.3×10^{-3}
0.2	0.229060	2.5319360	4.3×10^{-7}	2.5319363	8.8×10^{-6}	1.090071	1.5×10^{-7}	1.090070	2.8×10^{-3}
0.3	0.329212	2.6465810	1.2×10^{-6}	2.6465821	4.5×10^{-6}	1.207815	4.1×10^{-7}	1.207813	2.1×10^{-3}
0.4	0.422511	2.7651786	1.2×10^{-6}	2.7651797	4.6×10^{-6}	1.333843	4.3×10^{-7}	1.333842	2.7×10^{-3}
0.5	0.511244	2.8885340	8.0×10^{-7}	2.8885349	1.1×10^{-6}	1.444398	3.5×10^{-7}	1.444398	6.2×10^{-4}

Table 1: Extremal solutions constructed at $T = 0$ (here) against the manuscript’s $T \rightarrow 0$ extrapolation (ms), at the manuscript’s couplings ($g_H = 1$). “res.” is the larger of the differences between $N = 128$ and 96 and between 96 and 64. “spread” is the manuscript’s three-fit spread of μ and “envelope” its twenty-fit envelope for Ψ_{ext} .

Ψ_{ext} . To compare at the same couplings we refined our solutions at each of the manuscript’s 12 finite couplings with $N = 64, 96$ and 128 (table 1). Across all of them:

- y agrees to 3.4×10^{-7} ;
- μ agrees to 6.2×10^{-6} ;
- the slope agrees to 1.9×10^{-5} , which is at most 1.8×10^{-3} of the manuscript’s own envelope.

The two routes share no numerical machinery: different gauges, different discretisations, and an extrapolation in temperature against none. They agree far better than the manuscript’s uncertainty estimates suggest. Those estimates are therefore conservative. In a few cases the difference in μ exceeds the manuscript’s three-fit spread, but even then only at the level of 10^{-6} .

Evidence for the non-smooth horizon in the full solutions. The exponent γ comes from the linear analysis of section 3. The full solutions are consistent with it in three ways. The mass read from a second derivative converges only algebraically (figure 2), while an analytic solution would converge exponentially. The constraint violation peaks next to the horizon. So do the differences between resolutions. A direct fit of the decay of the Chebyshev coefficients does *not* resolve γ , however. The coefficients reach the round-off plateau by mode $n \approx 38$, before the algebraic decay $n^{-(4\gamma+1)}$ becomes visible. The full solutions support the non-smooth horizon but do not measure its exponent independently.

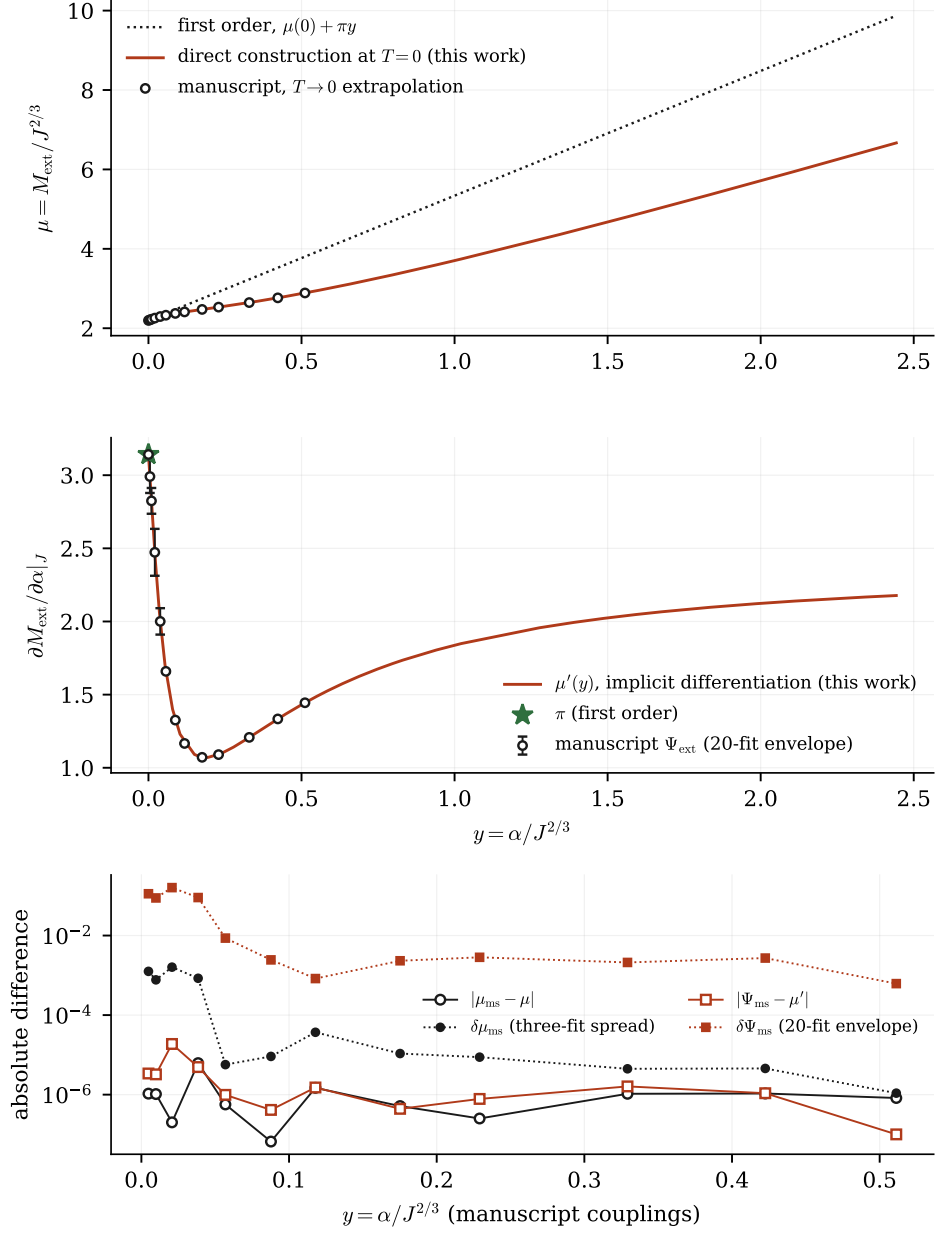


Figure 5: Top: extremal mass at fixed angular momentum from the direct construction (line), the manuscript’s $T \rightarrow 0$ extrapolations (circles), and the first-order result (dotted). Middle: slope from implicit differentiation (line), its first-order value π (star), and the manuscript’s Ψ_{ext} with its own uncertainty envelope (error bars). Bottom: the differences between the two routes (open symbols) compared with the manuscript’s own uncertainty estimates (filled symbols).

7 What was not established

- **The endpoint.** Our branch stops at $x = 0.863$ ($\alpha = 3.118$, $y = 2.443$). KKR reach $x \approx 0.92$. Their conjecture that the extremal set ends at the singular static point $(x, j) = (1, 0)$ is not tested. Extending the branch would need a coordinate that stretches the region next to the horizon, or extended-precision arithmetic, not more continuation steps.
- **Error bounds.** The resolution differences quoted throughout are differences, not bounds. The round-off floor grows with N and limits μ to about 10^{-7} – 10^{-6} .
- **Horizon regularity.** The non-integer exponent is established by the linear analysis only. We did not compute curvature in a parallelly propagated frame, so we do not claim a curvature singularity.
- **Field equations.** They were checked against the tensor equations on generic functions and against the closed-form static and Myers–Perry solutions. The near-horizon identification is KKR’s and was used as given.
- **KKR’s figure.** We compare with KKR’s curves only on the range we construct and make no statement about their accuracy beyond the figure. KKR’s scan of the interior of the domain of existence was not repeated. The interior points in figure 4 are the manuscript’s ($\alpha \leq \frac{1}{2}$).
- **Recalled, not derived.** The interpretation of the exponent pairing as an AdS_2 property, and the relation to [4], are recalled from the literature (the abstract of [4] was re-read for this write-up). The identities (9) and the slope formula (17) were derived here.

8 Summary

The extremal equal-spin black holes of five-dimensional Einstein–Gauss–Bonnet gravity can be built directly at zero temperature in KKR’s gauge. Doing so reliably needed three things beyond what KKR describe:

- removing two exact zero modes of the gauge (the size, and a leftover radial reparametrisation);
- a compactified coordinate in which the leftover gauge mode is smooth;
- unknowns in which the mass is a value at infinity rather than a derivative.

The last point matters because the extremal horizon carries a non-integer power ρ^γ at finite coupling, which slows the convergence of anything read from high derivatives.

With these solutions we reproduce KKR’s extremal curves to the precision of their figure. We also obtain the extremal mass and its slope at fixed angular momentum with no extrapolation in temperature. The slope falls from π to 1.0710 and rises again, but stays positive, so the extremality bound tightens with the coupling. Both the mass and the slope agree with the manuscript’s extrapolated values to a few parts in 10^6 : an independent confirmation of the manuscript’s central numbers, and evidence that its quoted uncertainties are conservative.

A Reproducing the results

All files are in `work/kkr-extremal`. Use the project environment (`.venv`: Python 3.10.12, NumPy 2.2.6, SciPy 1.15.3, SymPy 1.14.0). In order:

command	produces
<code>python derive.py</code>	field equations with g free, checked against the tensor route
<code>python nearhorizon.py</code>	throat, KKR closed forms, exponents $\gamma(y)$
<code>python kkr_equations.py 0</code>	residuals in KKR's gauge as NumPy code (~ 7 min)
<code>python pw_generator.py 0</code>	angular momentum from the conserved momentum
<code>python test_rq.py</code>	solver gate and resolution study
<code>python branch_rq.py ...</code>	the extremal branch (several phases, resumable)
<code>python polish_couplings.py</code>	high-resolution solutions at the manuscript's couplings
<code>python eps_digitise.py;</code> <code>python kkr_curves.py</code>	KKR's curves from their EPS files
<code>python analyse_branch.py</code>	checks and comparisons
<code>python -m marimo export html</code> <code>figures_kkr.py -o figures_kkr.html</code>	figures
<code>python make_report_numbers.py</code>	every number quoted in this report

The arXiv source of [1] (needed for the EPS files) is not redistributed. It is fetched from <https://arxiv.org/e-print/2303.12471> into `arxiv/`. The original computation ran on 7 October 2026: the branch from 09:47 to 14:10, and the refined solutions from 10:29 to 10:46. `RESUME.txt` records the resume commands and the dead ends in more detail.

Receipt for this write-up (2026-10-07, 16:03–16:07). The figure notebook was re-run (`python -m marimo export html figures_kkr.py`), which added figures 3 and 2. The solver was re-run from scratch as a spot check (extremal Myers–Perry start, α in steps of 0.01 to 0.2, $N = 64$, 245 s). Output:

```
alpha=0: M=2.356194490192 (3pi/4=2.356194490192) J=1.110720734540
(pi sqrt(2/4)=1.110720734540) Omega_H=0.707106781187
alpha=0.010 mu=2.22627827 ms=2.22627878 d=5.1e-07 | mu'=2.8245634 Psi_ms=2.8245576
alpha=0.020 mu=2.25505852 ms=2.25505786 d=6.6e-07 | mu'=2.4728489 Psi_ms=2.4728269
alpha=0.050 mu=2.32801712 ms=2.32801721 d=9.7e-08 | mu'=1.6590803 Psi_ms=1.6590800
alpha=0.100 mu=2.41072791 ms=2.41072869 d=7.8e-07 | mu'=1.1662206 Psi_ms=1.1662137
alpha=0.150 mu=2.47399296 ms=2.47399319 d=2.3e-07 | mu'=1.0714928 Psi_ms=1.0714924
alpha=0.200 mu=2.53193599 ms=2.53193626 d=2.7e-07 | mu'=1.0900711 Psi_ms=1.0900703
```

The branch, the refined solutions and the near-horizon scan were not re-run for this write-up. Their numbers are taken from the stored outputs through `./report/numbers.tex`.

References

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